HYBRID PARTICIPATION INNOVATION LAB · STEP 7 · 18 NOVEMBER

# The citizen science component

"Energy Detectives": tenants produce the knowledge the AI is missing

**Next Wave**

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RESEARCH QUESTIONS

# Three things we need to know

**RQ1** When do households really use energy, and what stops them from using it in sunny hours?

**RQ2** Where does the AI forecast go wrong, and for which kinds of households?

**RQ3** How would each fairness rule change what real households receive?

THE SIX-PART DESIGN

# From question to change

| Part | Our answer |
|---|---|
| **Question** | RQ1–RQ3: routines, forecast errors, effects of fairness rules |
| **Protocol** | Co-written in a workshop by tenants and researchers; tested with 5 households first |
| **Participation** | Postcards for all; smart plugs, diaries and shade photos for volunteers |
| **Validation** | Diaries checked against meter data; training; uncertainty shown openly |
| **Interpretation** | Monthly data evenings where tenants and researchers read results together |
| **Feedback** | Findings change the model, the messages and the fairness rule, logged publicly |

WHAT CITIZENS ACTUALLY DO

# Observe, measure, document, classify, interpret

### WIDE TRACK · ALL HOUSEHOLDS

**Document:** a two-week energy postcard, ticking when big appliances run.

**Classify:** "could I have moved this to midday? yes / no / why".

**Consent:** optional sharing of 15-minute meter data.

### DEEP TRACK · ~30 VOLUNTEERS

**Measure:** smart plugs on the washing machine or water heater.

**Observe:** photos of roof shade at fixed times.

**Interpret:** label data, compare it with the AI forecast, propose changes.

DATA QUALITY AND UNCERTAINTY

# Credible enough for ATER and GSE to listen

### Who defines the protocol

Tenants and researchers together, then a 5-household pilot to fix unclear steps.

### Cross-checks

Postcards and diaries compared with meter data; mismatches discussed, not deleted.

### Training and support

A 30-minute session, a picture guide and an ambassador to call.

### Honest uncertainty

Every chart shows how sure we are; small samples are labelled as such.

AVOIDING THE EXTRACTIVE MODEL

# Tenants are not unpaid sensors

### Data belongs to the CER

The provider only processes it, under the tenants' data charter.

### Privacy by design

Aggregated by default, deleted after the pilot, GDPR consent per use.

### Opt out, no penalty

Not sharing data never reduces a household's share.

### Results returned

Each household gets its own "energy postcard" in plain language.

### Right to challenge

Any finding can be contested at a data evening and re-checked.

### Recognition

Named credit, certificates and a small allowance for volunteers.

HOW KNOWLEDGE ENTERS DECISIONS

# From a postcard to a change in the AI

**1 · DATA**

Postcards, plugs and photos collected

→

**2 · DATA EVENING**

Tenants and researchers interpret together

→

**3 · PROPOSAL**

A written change request with evidence

→

**4 · DECISION**

Assembly or provider must answer within a month

→

**5 · CHANGE LOG**

Public record: accepted, changed, or refused with reasons

**EXAMPLE**

Diaries show shift workers can't shift laundry to noon → the assembly adds a "flexibility credit" for night-time water-heater timers → the AI learns a new household type.